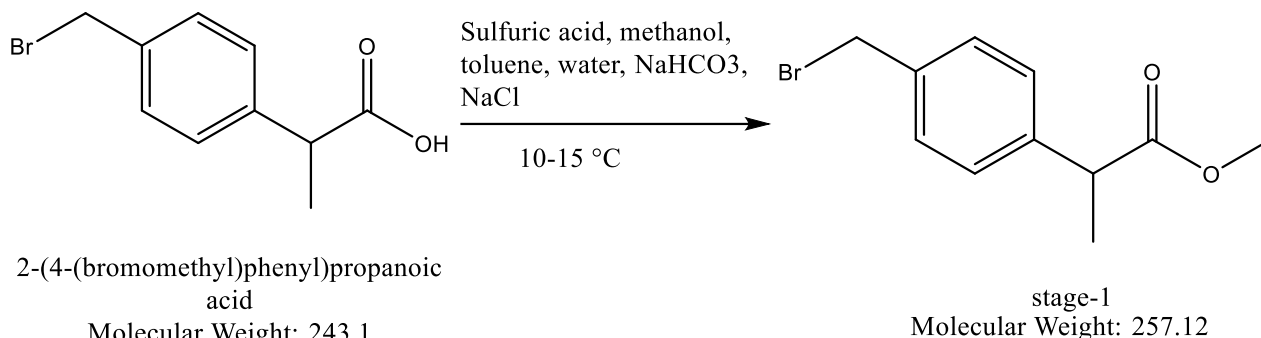


PROCESS WRITE-UP
FOR
LOXOPROFEN SODIUM STAGE 1

	Name	Designation	Sign and Date
Prepared by CRD			
Checked by CRD			
Approved by DQA			

Synthesis of Stage 1 (LOXOPROFEN SODIUM):**Raw Materials:**

1	2-(4-(bromomethyl)phenyl)propanoic acid	100.0	g	243.1	0.411	1.0
2	Sulfuric acid	40.35	g	98.08	0.411	1.0
3	Methanol	300.0	mL	-	-	3
4	Toluene Lot-1	200.0	mL	-	-	2
5	Toluene Lot-2	200.0	mL	-	-	2
6	Toluene Lot-3	200.0	mL	-	-	2
7	Water Lot-1	200.0	mL	-	-	2
8	Water Lot-2	200.0	mL	-	-	2
9	Water Lot -3	100.0	mL	-	-	1
10	Sodium bi carbonate Lot-1	10	g	84	0.119	0.3
11	Sodium bi carbonate Lot-2	10	g	84	0.119	0.3
12	Sodium chloride	5	g	58.44	0.085	0.21
13	Sodium sulphate	5	g	142.04	0.035	0.1

Procedure:

1. Arrange clean and dry RBF.
2. Charge Methanol under stirring into the RBF at 25-30°C.
3. Charge 2-(4-(bromomethyl) phenyl) propanoic acid at 25-30°C.
4. Cool the reaction mass temperature to 5-15°C.
5. Slowly add Sulfuric acid into the reaction mixture at 5-15°C, Exotherm observed.
6. Maintain the reaction mass at 5-15°C for 3 hours. & submitted the **IPC-1** sample for checking 2-(4-(bromomethyl) phenyl) propanoic acid content by HPLC (**Limit: NMT 0.5%**), if result complies follow

Step No:7 and if result does not comply, follow the next Step No:6 and submit the sample for every 1 hour.

7. Charge Toluene Lot-1 & stir the reaction mass for 15 minutes at 10-15°C.
 8. Meanwhile prepare Sodium bicarbonate solution Lot-1 (Sodium bicarbonate Lot-1 in Water Lot-1)
 9. Slowly add the Sodium bicarbonate solution in reaction mass at 10-20°C.
 10. Stir the reaction mass for 10-15 minutes at 25-30°C.
 11. Settle and separate both the layers for 30 minutes
 12. Charge the Aqueous layer into the RBF at 25-30°C. Organic Layer.....Qty
 13. Add Toluene Lot-2 to the above Aqueous layer at 25-30°C.
 14. Stir the reaction mass for 10-15 minutes at 25-30°C.
 15. Settle and separate both the layers for 10-15 minutes
 16. Charge the Aqueous layer into the RBF at 25-30°C. Organic Layer.....Qty
 17. Add Toluene Lot-3 to the above Aqueous layer at 25-30°C.
 18. Stir the reaction mass for 10-15 minutes at 25-30°C.
 19. Settle and separate both the layers for 10-15 minutes.
 20. Charge the Aqueous layer into the RBF at 25-30°C. Organic Layer.....Qty
 21. Charge all the Organic layers from Step. No.12,16,20 into the RBF at 25-30°C.
 22. Meanwhile prepare Sodium bicarbonate solution Lot-2 (Sodium bicarbonate Lot-2 in Water Lot-2)
 23. Charge Sodium bicarbonate solution Lot-2 to the above Organic layers at 25-30°C.
 24. Stir the reaction mass for 10-15 minutes at 25-30°C
 25. Settle and separate both the layers for 10-15 minutes.
 26. Charge the Aqueous layer into the RBF at 25-30°C. Organic Layer.....Qty
 27. Meanwhile prepare Sodium Chloride solution (Sodium Chloride in Water Lot-3)
 28. Charge the Organic layer from Step No.25 into the RBF at 25-30°C.
 29. Charge Sodium chloride solution in Organic layer 25-30°C.
 30. Stir the reaction mass for 10-15 minutes at 25-35°C.
 31. Settle and separate both the layers for 10-15 minutes.
 32. Charge the Aqueous layer into the RBF at 10-15°C. Organic Layer.....Qty
 33. Collected Organic layer and dry with Sodium Sulphate at 25-35°C.
 34. Distill off toluene completely under vacuum at below 50°C.
- Note:** - Don't go more temperature while distillation due to impurities formation at higher temperature.
35. Cool the reaction mass temperature to 30-35 °C.

36. Unload the crude and record the weight.

Quantity.....gm

37. Submit sample for complete analysis.

Input (g)	Output (g)	Expected Yield (w/w)	Theoretical Yield (w/w)	% Yield
100	100-110	1.0-1.1	1:1.05	95-104